

Question 4

Online Shopping Bill Calculator.
PAC Chart

Given Data

- Quantity q
- Price per item p
- discount d
- Tax t

Required Result

- Final bill
- Bill document
- Error message for invalid

Process Required

- Input q, p, d, t
- Validate all inputs
- Calculate $s = q \times p$
- Calculate discounted amount
- Calculate final bill
- Store the details and generate bill
- Display final bill

Alternative Solution

- Use separate functions for each calculations
- Use one function to calculate complete bill.

Online Shopping Bill Calculator Algorithm

1. Start
2. Take input q, p, d, t from the user.
3. Check for all input values wheather they are valid.
4. if any value is invalid, display an error message and terminate the program.
5. Calculate the subtotal, $subtotal = q \times p$
6. Calculate the discounted-amount.
 - $discounted_amount = s - (s * d) / 100$
7. Calculate the final-Bill
 - $final_Bill = discounted_amount + (discounted_amount * t) / 100$
8. Generate the Bill
9. Display the subtotal, discounted-amount and final-Bill.
10. Stop.

Online Shopping Bill Calculator

IPO Chart

Input	Process	Output
- Quantity q	Validate all inputs	Error message if invalid
- Price per item p	Calculate subtotal. $S = q \times p$	Subtotal
- Discount d	Calculate discounted amount: $as = S - (S \times d)$	Discounted amount
- Tax % t	Calculate final Bill. $Final = as \times (1 + t) / 100$	Final Bill
-	Display bill	

Online Shopping Bill Calculator

Pseudocode

Begin

Declare q as Integer and p, d, t, s, a
and final_Bill as float

input q input p input d input t

if $q \leq 0$ OR $p \leq 0$ OR $d \leq 0$ OR $t \leq 0$
Display "Invalid input. Please enter valid values".

end if

subtotal $\leftarrow q \times p$ discounted_amount $\leftarrow s - (s \times d) / 100$ final_Bill $\leftarrow \frac{\text{discounted_amount} + (\text{discounted_amount} \times t)}{100}$

Display "Shopping Bill"

Display subtotal

Display discounted-amount

Display final_Bill

End

Flowchart

